arXiv:2401.18000

# Multi-Agent Reinforcement Learning for Strategic Board Games with Imperfect Information

Linfeng Zhao, Daria de Tinguy, Tomoya Tanaka

Year: 2024    Venue: AAAI Conference on Artificial Intelligence

## Abstract

We extend AlphaZero-style self-play to imperfect-information board games such as Stratego, Diplomacy, and Hanabi. Our agent combines counterfactual regret minimization with a learned value network and reaches expert-level play within 18 hours of TPU training. The method is purely a discrete-state turn-based game-AI study and does not involve any robotic embodiment, physical actuators, sensorimotor policies, or sim-to-real transfer.

## Introduction

This paper (arXiv:2401.18000) is titled 'Multi-Agent Reinforcement Learning for Strategic Board Games with Imperfect Information'. The work was released in 2024. The abstract reads: We extend AlphaZero-style self-play to imperfect-information board games such as Stratego, Diplomacy, and Hanabi. Our agent combines counterfactual regret minimization with a learned value network and reaches expert-level play within 18 hours of TPU training. The method is purely a discrete-state turn-based game-AI study and does not involve any robotic embodiment, physical actuators, sensorimotor policies, or sim-to-real transfer. We expand on the motivation, prior art, and the contribution claims of the work in the remainder of this section. The narrative below is intentionally synthesized to match the style of an arXiv preprint while preserving the core technical claim of the abstract above.

## Related Work

We compare against prior work spanning the relevant subfield. For arXiv:2401.18000, the closest baselines are the ones cited in the original report. We discuss contemporaneous work and explain how the present submission differentiates itself in terms of method, evaluation, and scope.

## Method

Our approach for Multi-Agent Reinforcement Learning for Strategic Board Games with Imperfect Information follows the abstract: We extend AlphaZero-style self-play to imperfect-information board games such as Stratego, Diplomacy, and Hanabi. Our agent combines counterfactual regret minimization with a learned value network and reaches expert-level play within 18 hours of TPU training. The method is purely a discrete-state turn-based game-AI study and does not involve any robotic embodiment, physical actuators, sensorimotor policies, or sim-to-real transfer. Implementation details, hyperparameters, and the loss construction are documented here for reproducibility.

## Experiments

Experimental results validate the central claim. Tables and figures (omitted in this placeholder render) report the headline numbers cited in the abstract: We extend AlphaZero-style self-play to imperfect-information board games such as Stratego, Diplomacy, and Hanabi. Our agent combines counterfactual regret minimization with a learned value network and reaches expert-level play within 18 hours of TPU training. The method is purely a discrete-state turn-based game-AI study and does not involve any robotic embodiment, physical actuators, sensorimotor policies, or sim-to-real transfer.

## Discussion

Limitations of the present approach are discussed. We note where the method may fail to generalize and propose extensions that future work should address.

## Conclusion

We conclude by summarizing the contribution of arXiv:2401.18000 and outline directions for follow-up work.

## Introduction

This paper (arXiv:2401.18000) is titled 'Multi-Agent Reinforcement Learning for Strategic Board Games with Imperfect Information'. The work was released in 2024. The abstract reads: We extend AlphaZero-style self-play to imperfect-information board games such as Stratego, Diplomacy, and Hanabi. Our agent combines counterfactual regret minimization with a learned value network and reaches expert-level play within 18 hours of TPU training. The method is purely a discrete-state turn-based game-AI study and does not involve any robotic embodiment, physical actuators, sensorimotor policies, or sim-to-real transfer. We expand on the motivation, prior art, and the contribution claims of the work in the remainder of this section. The narrative below is intentionally synthesized to match the style of an arXiv preprint while preserving the core technical claim of the abstract above.

## Related Work

We compare against prior work spanning the relevant subfield. For arXiv:2401.18000, the closest baselines are the ones cited in the original report. We discuss contemporaneous work and explain how the present submission differentiates itself in terms of method, evaluation, and scope.

## Method

Our approach for Multi-Agent Reinforcement Learning for Strategic Board Games with Imperfect Information follows the abstract: We extend AlphaZero-style self-play to imperfect-information board games such as Stratego, Diplomacy, and Hanabi. Our agent combines counterfactual regret minimization with a learned value network and reaches expert-level play within 18 hours of TPU training. The method is purely a discrete-state turn-based game-AI study and does not involve any robotic embodiment, physical actuators, sensorimotor policies, or sim-to-real transfer. Implementation details, hyperparameters, and the loss construction are documented here for reproducibility.

## Experiments

Experimental results validate the central claim. Tables and figures (omitted in this placeholder render) report the headline numbers cited in the abstract: We extend AlphaZero-style self-play to imperfect-information board games such as Stratego, Diplomacy, and Hanabi. Our agent combines counterfactual regret minimization with a learned value network and reaches expert-level play within 18 hours of TPU training. The method is purely a discrete-state turn-based game-AI study and does not involve any robotic embodiment, physical actuators, sensorimotor policies, or sim-to-real transfer.

## Discussion

Limitations of the present approach are discussed. We note where the method may fail to generalize and propose extensions that future work should address.

## Conclusion

We conclude by summarizing the contribution of arXiv:2401.18000 and outline directions for follow-up work.

## References

[1] Smith, J. et al. Sample reference for padding the PDF length. Journal of Placeholder Studies, 2024.

[2] Smith, J. et al. Sample reference for padding the PDF length. Journal of Placeholder Studies, 2024.

[3] Smith, J. et al. Sample reference for padding the PDF length. Journal of Placeholder Studies, 2024.

[4] Smith, J. et al. Sample reference for padding the PDF length. Journal of Placeholder Studies, 2024.

[5] Smith, J. et al. Sample reference for padding the PDF length. Journal of Placeholder Studies, 2024.

[6] Smith, J. et al. Sample reference for padding the PDF length. Journal of Placeholder Studies, 2024.

[7] Smith, J. et al. Sample reference for padding the PDF length. Journal of Placeholder Studies, 2024.

[8] Smith, J. et al. Sample reference for padding the PDF length. Journal of Placeholder Studies, 2024.

[9] Smith, J. et al. Sample reference for padding the PDF length. Journal of Placeholder Studies, 2024.

[10] Smith, J. et al. Sample reference for padding the PDF length. Journal of Placeholder Studies, 2024.